

HW3:

R3)

Forwarding and routing are two distinct functions performed at the network layer. Forwarding is a local router action that involves moving a packet from an input link interface to the appropriate output link interface. This is a hardware-level process that happens in very short timeframes. Routing is a network-wide process that determines the end-to-end path packets take from a source host to a destination host. Routing involves the coordination of all routers in a network through the use of routing protocols and algorithms to calculate these paths.

R17)

When Host B receives an IP datagram, its network layer needs to figure out which upper-layer protocol should receive the encapsulated segment. It does this by looking at a specific 8-bit field in the IP datagram header known as the protocol field. This field contains a specific value that indicates whether the payload is meant for TCP, UDP, or another protocol.

P1a)

Host h3 → 3

P1b)

It is not possible to write a standard forwarding table that sends H1's traffic to interface 3 and H2's traffic to interface 4 if they are both going to H3. This is because traditional forwarding in the network layer is destination-based. The router only looks at the destination IP address to decide where to send the packet, and since the destination for both is H3, they have to follow the same path from router A.

P8a)

11100000 00 → Interface 0

11100000 01000000 → Interface 1

11100000 01 → Interface 2

11100001 01 → Interface 2

Otherwise → Interface 3

P8b)

11001000 10010001 01010001 01010101 → Interface 3. This address doesn't match any of the top four prefixes, so it defaults to the "otherwise" route.

11100001 01000000 11000011 00111100 → Interface 2. The first 10 bits (11100001 01) match the fourth entry in our table.

11100001 10000000 00010001 01110111 → Interface 3. Even though it starts with 11100001, the 9th and 10th bits are "10" which doesn't match our Interface 2 prefix, so it falls to "otherwise."

P9)

Interface 0 → Binary 00000000 to 00111111 → Decimal 0 to 63 → 64 addresses

Interface 1 → Binary 01000000 to 01111111 → Decimal 64 to 127 → 64 addresses

Interface 2 → Binary 10000000 to 10111111 → Decimal 128 to 191 → 64 addresses

Interface 3 → Binary 11000000 to 11111111 → Decimal 192 to 255 → 64 addresses

P14)

An example of one IP address that can be assigned to this network is 128.119.40.129.

The prefixes for the four subnets are:

128.119.40.64/28

128.119.40.80/28

128.119.40.96/28

128.119.40.112/28

P16)

The IP address range for Gonzaga is 147.222.0.0 to 147.222.255.255

P20)

Match: Ingress Port 1, Source IP 10.3.0.5 or 10.3.0.6, Destination IP 10.1.0.1 or 10.1.0.2 Action: Forward to port 2

Match: Ingress Port 2, Source IP 10.1.0.1 or 10.1.0.2, Destination IP 10.3.0.5 or 10.3.0.6 Action: Forward to port 1

Match: Destination IP 10.2.0.3 Action: Forward to port 3

Match: Destination IP 10.2.0.4 Action: Forward to port 4